

Nicolas Mir

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SUMMARY

Data Scientist with a dual background in Data Science (M.S., University of Auckland, 2025) and Software Engineering (MEng, ENSIIE, France). Hands-on experience building predictive models, end-to-end data pipelines, and interactive dashboards from large, multi-source datasets. Proven ability to translate statistical findings into clear, actionable insights for non-technical stakeholders. Passionate about applying machine learning and hypothesis-testing skills to real-world operational problems.

TECHNICAL SKILLS

Core Languages: Python, R, Bash, SQL

Data & ML Libraries: NumPy, Pandas, Scikit-learn, PyTorch, Tensorflow, NLTK

Statistical Methods: Hypothesis testing, time-series forecasting, sequence models, regression, clustering (GMM)

Databases: SQL (MySQL, PostgreSQL), NoSQL (Neo4j), Snowflake, dbt, ETL/ELT pipelines

Data Visualisation: Power BI (DAX), Matplotlib, ggplot2 (R), Streamlit

Cloud & Big Data: Microsoft Azure (DP-900 certified), Spark, Hadoop, Docker

Dev Tools: Git, Prefect (orchestration), CI/CD, Excel

SELECTED PROJECTS

Unemployment prediction in New Zealand

2026

Trend prediction

DAX, Power BI, python, Multi-source Data Integration

- Built an end-to-end analytics pipeline integrating multi-source economic data (Stats NZ, RBNZ, World Bank) to model national unemployment trends and regional disparities (1999–2022).
- Developed interactive Power BI dashboards surfacing macro-economic relationships (OCR, CPI, GDP) and regional inequality – delivering complex insights accessibly to non-technical audiences.

Large-Scale ELT Pipeline – Global YouTube Trending Analysis

2026

Preprocessing and Multimodal

Snowflake, ELT, SQL, dbt, CI/CD, Streamlit

- Engineered a production-grade ELT pipeline ingesting live data into Snowflake, transforming with dbt, and orchestrating daily workflows via Prefect – demonstrating ability to handle large-scale operational datasets reliably in production.
- Delivered an interactive Streamlit dashboard for engagement-metric analysis on a fully free-tier, cost-efficient cloud architecture.

Cloud forecasting research

2025

Computer Vision Project

Python, Stable Diffusion Model, CVAT, Quantitative Evaluation (PSNR, RMSE, SSIM)

- Designed and trained a generative sequence model to forecast cloud motion 10 minutes ahead – directly applicable to predictive failure and early-warning signal modelling.
- Rigorously evaluated model performance using RMSE and SSIM; documented findings in a 50-page thesis, demonstrating scientific reporting and communication skills.

EXPERIENCE

API Data Developer

Sep 2023 – Feb 2024

Altermaker

Troyes, France

- Designed a 30-table MySQL relational database and REST API; contributed to a client-facing product serving 1,000+ users.
- Gained end-to-end experience across database design, API development, and data integration.

Research Intern

May 2021 – Nov 2021

LIRIS Laboratory

Lyon, France

- Conceived and implemented a GMM-based statistical method for automated graph schema discovery in Big Data, outperforming prior benchmarks in both speed and precision (Python, Neo4j)..
- Co-authored a peer-reviewed publication: Hierarchical Clustering for Property Graph Schema Discovery – demonstrating hypothesis-driven research and scientific communication..

Software Tester and Developer
Stafiz

May 2022 – Nov 2022
Lisbon, Portugal

- Built automated test suites (Laravel Dusk) to protect critical production features; worked with REST APIs and AWS infrastructure.

EDUCATION

University of Auckland
M.S. Data Science

New Zealand
Mar 2024 – Dec 2025

ENSIIE
Software Engineering Degree

Évry, France
Sep 2019 – Dec 2022

Preparatory Classes MPSI (MPSI – Mathematics & Physics)

Troyes, France
Sep 2017 – Jul 2019

CERTIFICATIONS

- Azure Data Fundamentals Certification (DP-900)
- Coursera: SQL at Scale: Querying, Transforming, and Governing (window functions, CTEs, UNPIVOT, etc.)
- Udemy: AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents

LANGUAGES

- French - native
- English - professional working proficiency
- Russian - B2
- German - B1

HOBBIES

- Piano at the conservatory (13 years)
- Competitive chess (2000 FIDE)
- Running and squash